

GIVELIFE

Organ Donor Registry Portal

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Abstract

The GiveLife — Organ Donor Registry Portal is a web-based application developed to simplify the process of organ donor registration and basic donor-recipient matching. The main aim of this project is to create an easy-to-use platform where individuals can register as organ donors, provide their consent, and view relevant donor information in a clear and organized manner.

The system includes key features such as a donor registration form, a consent form where users officially agree to donate their organs, a donor list display, and a basic matching system. The matching process is implemented using JavaScript by comparing important details such as blood group and organ type. The application also provides simple status tracking, indicating whether a donor is available or matched.



Problem Statement

Low Awareness

Millions of people are unaware of the organ donation process, consent requirements, or how to register as a donor.

No Unified Platform

There is no simple, accessible portal where individuals can register, provide consent, and view donor-recipient match information in one place.

Complex Systems

Existing solutions involve heavy backend infrastructure, making them unsuitable for learning, community outreach, or low-resource environments.

Proposed System

GiveLife proposes a fully client-side web portal that eliminates backend complexity while delivering a complete organ donor registration experience — from signup to donor-recipient matching.

Donor Registration

Users fill a detailed form capturing name, blood group, organ type, and contact info.

Consent Form

Donors digitally agree to donation terms through a dedicated consent workflow.

Donor List

All registered donors are displayed in an organized, filterable list view.

Match & Status

JavaScript matches donors to recipients by blood group and organ type, with live status.

Modules Covered in the Project

1. Donor Registration Module

This module handles the process of registering new organ donors into the system. Users provide personal details such as their full name, age, gender, blood group, organs they wish to donate, phone number, and city. Input validation ensures that only eligible donors (aged 18–65) are accepted. Upon successful registration, each donor is assigned a unique Donor ID (e.g., GL83921) and their record is saved to local storage. The module also includes a form reset feature to clear all fields after submission.

2. Consent Management Module

This module manages the legal and ethical consent process for organ donation. Donors are presented with a detailed consent agreement that outlines their rights, the conditions of donation, and the limitations of the process. Users must confirm their agreement by checking two declaration checkboxes and providing their name and date. An optional field allows donors to link their consent to an existing Donor ID. Submitted consent records are confirmed via a toast notification, ensuring transparency and accountability throughout the process.

3. Donor Registry & Search Module

This module displays all registered donors in a searchable, filterable table. Administrators or medical staff can search donors by name, city, blood group, or organ type. Filters allow narrowing results by blood group and availability status (Available or Matched). Each donor record shows their ID, name, age, blood group, organ, city, and current status. Actions include marking a donor as Matched or removing them from the registry. The table dynamically updates based on search and filter inputs without any page reload.

4. Donor Matching Module

This module enables medical personnel to find a compatible organ donor for a recipient. The user selects the recipient's blood group and the organ required. The system applies blood compatibility rules (e.g., O- is a universal donor; AB+ can receive from all groups) to identify available donors whose organ and blood group match the recipient's needs. If a compatible donor is found, their details are displayed and staff can confirm the match, which updates the donor's status to Matched across the entire registry.

5. Notification & Stats Module

This module keeps users informed through real-time feedback and live statistics. A toast notification system displays success or error messages after every key action — registration, consent submission, matching, or deletion. The home page stats bar dynamically reflects the total number of registered donors, how many are currently available, and how many have been matched. These figures update automatically whenever any change is made to the registry, giving a live snapshot of the portal's activity at all times.

Thank You

